

Intelligent Instrument Cooling Fan Driver



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The circuit described here is of a temperature-controlled driver for cooling an instrument fan. The circuit operates off a single 12V DC supply voltage. None of the component values are especially critical.

Cooling fans can be necessary for circuit cooling but continually running a cooling fan at the rated voltage can drain the power supply

and shorten its lifetime. By running the cooling fan only as per requirement, you can extend its useful life as well as the life of the instrument under cooling.

Circuit and working

Fig. 1 shows the circuit diagram of the cooling fan driver. It is divided into two parts: fan driver circuit and fan status monitor circuit.

The fan driver circuit is shown on the left side of the dotted line in Fig. 1. It is built around a thermistor (NTC1) with negative temperature coefficient (NTC), N-channel MOSFET 60NF06 (IRF1) and a few other components.

NTC1 is used as a temperature

sensor, and IRF1 is used for driving the cooling fan. At normal temperature, resistance of the thermistor is very high ($> 9k$), and gate voltage of the MOSFET (IRF1) is well below its typical threshold value $V_{gs(th)}$; so, the cooling fan is off.

As the temperature rises, resistance of the thermistor drops ($< 6k$), causing the gate voltage to rise above $V_{gs(th)}$, and IRF1 begins to conduct. At still higher temperatures, IRF1 saturates and the cooling fan runs at its maximum speed. The 10k potentiometer (VR1) determines the threshold temperature at which the cooling fan starts running.

N-channel MOSFET IRF1 has a gate threshold with a negative temperature coefficient, which helps to relieve the effects of possible $V_{gs(th)}$ discrepancy.

60NF06 MOSFET used here is a fast switching device in TO-220 package. It has a very low RDS(on) of 0.016 Ω max.

If you want only a fan driver circuit, the fan status monitor circuit

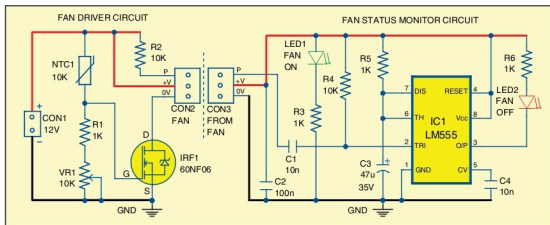


Fig. 1: Circuit of the cooling fan driver

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